آغا خان یونیورسٹی ایگزامینیشن بورڈ

AGA KHAN UNIVERSITY EXAMINATION BOARD

Notes from E-Marking Centre on HSSC-II Biology Annual Examinations 2025

Introduction

This document has been produced for the teachers and candidates of Higher Secondary School Certificate (HSSC) Part II Biology. It contains comments on candidates' responses to the 2025 HSSC-II Examination indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses that support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. It is imperative to refer to command word guide available on AKU-EB website for understanding the expectations of the command word.

General Observations

Overall, candidate performance was encouraging, with many demonstrating strong understanding of key biological concepts. Most outperformed in areas such as mineral homeostasis, sympathetic nervous system, Follicle Stimulating Hormone (FSH) and Luteinising Hormone (LH) functions, skin ageing factors, metastatic cells, and reflex arc, reflecting solid grasp of both foundational and applied content. Nonetheless, it is important for teachers to place greater emphasis on specific areas where candidates showed weaker understanding.

Topics such as:

- exemplification of positive feedback mechanism in biological systems.
- types of epistasis.
- the termination phase of translation in the process of protein synthesis.
- types of vectors used in gene therapy.
- osmoregulatory adaptations in freshwater bony fish.

- causes and treatment of slipped disc.

Note: Candidates' responses shown in this report have not been corrected for grammar, spelling, format, or information.

DETAILED COMMENTS
Constructed Response Questions (CRQs)

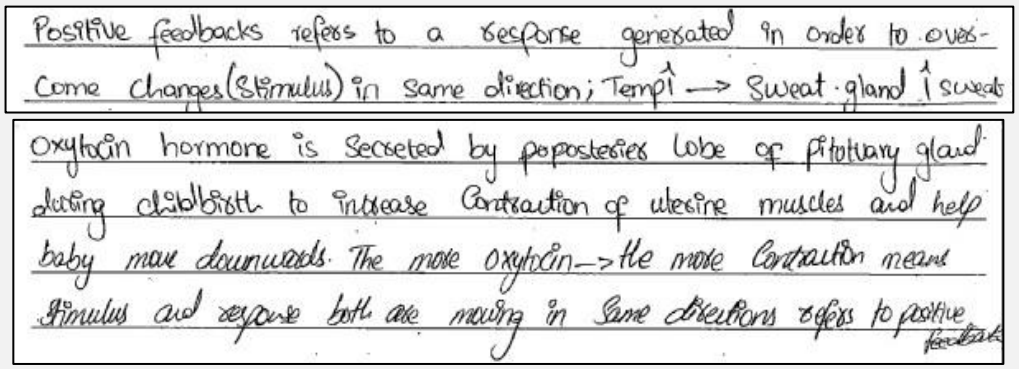
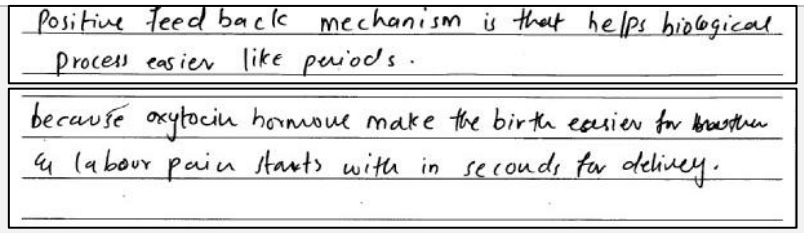
Question No. 1	
Question Text	a. Define positive feedback mechanism in biological systems. b. Why is the production of oxytocin hormone during childbirth considered a positive feedback mechanism?
SLO No.	15.2.3
SLO Text	Compare positive and negative feedback with examples.
Max Marks	3
Cognitive Level	U*
Checking Hints	a. 1 mark for definition b. 1 mark for recognising that oxytocin stimulates uterine contractions during childbirth 1 mark for describing that as contractions intensify, oxytocin release increases further
Overall Performance	Overall, candidates demonstrated an average understanding of the positive feedback mechanism. While many could identify oxytocin as an example, they often struggled to clearly explain how it reinforces the cycle during childbirth and to distinguish it from negative feedback.
Description of Better Responses	*Better responses* reflected a clear understanding of positive feedback, accurately defining it as a regulatory process in which the response amplifies the initial stimulus, increasing deviation from the original state. These candidates effectively used the example of oxytocin, explaining that its release from the posterior pituitary during childbirth increases uterine contractions, which in turn stimulates further oxytocin release. This correctly illustrates the self-reinforcing nature of positive feedback, distinguishing it from negative feedback mechanisms.
Image of Better Response	
Description of Weaker Responses	*Weaker responses* struggled to define the positive feedback mechanism, with some incorrectly describing a negative feedback process instead. Many candidates misunderstood the question, which specifically asked why oxytocin production during childbirth is considered a positive feedback mechanism. Instead of addressing this, they explained oxytocin's role in milk production during lactation, which was not relevant. This indicates a lack of understanding of the feedback concept and an inability to interpret the question accurately.

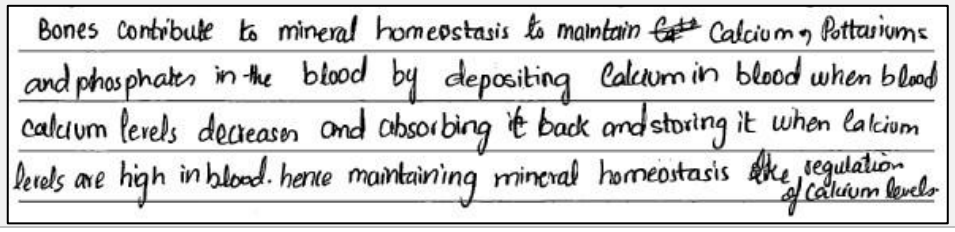
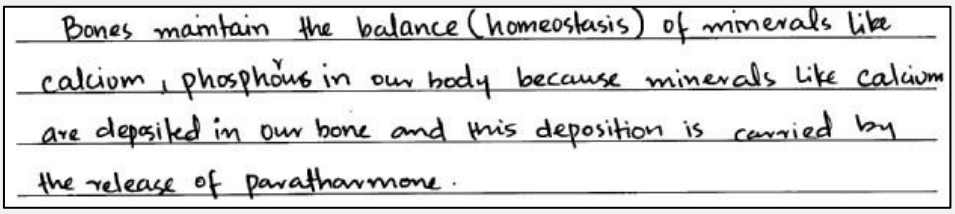
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	**Preferred Pedagogy** Used for this SLO**	**Assessment Strategies**
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration ** For description of each Pedagogy, refer to Annexure A	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion: Use comparative examples from human physiology, such as childbirth (positive) and thermoregulation or blood glucose regulation (negative) to distinguish between positive and negative feedback mechanisms. Visual aids like flowcharts and diagrams can help students understand the self-amplifying nature of positive feedback. Incorporating concept-check questions and classroom discussions focused on real-life scenarios will encourage critical thinking and help students better interpret exam questions.

*K = Knowledge U = Understanding A = Application and other higher-order cognitive skills

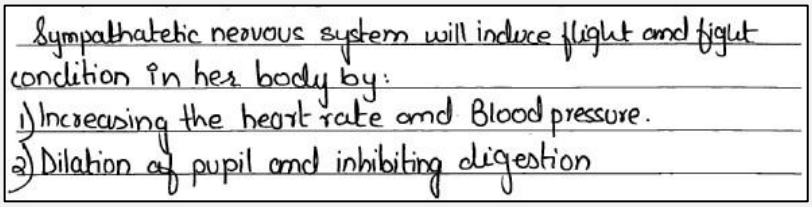
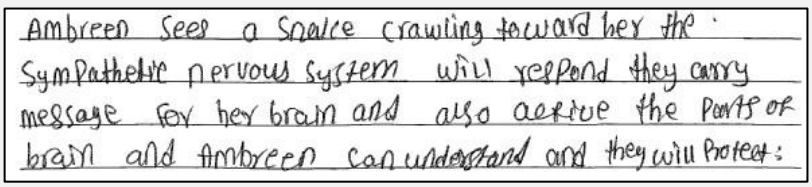
Question No. 2	
Question Text	Briefly describe how bones contribute to mineral homeostasis, taking calcium as an example.
SLO No.	16.4.3
SLO Text	Discuss functions of human skeleton.
Max Marks	2
Cognitive Level	U
Checking Hints	1 mark for deposition of calcium by bones 1 mark for release of calcium (demineralisation)
Overall Performance	Most candidates demonstrated a sound understanding of the role of bones in mineral homeostasis. The use of relevant terminology and clear, structured answers indicated good conceptual understanding and effective preparation.
Description of Better Responses	*Better responses* accurately explained that bones act as a storage site for calcium, highlighting that calcium is deposited in bones when blood calcium levels are high due to sufficient dietary intake. They also correctly described that, during low calcium intake, bones release calcium through demineralisation to maintain blood calcium levels. These candidates demonstrated a clear understanding of the dynamic role of bones in regulating calcium balance.
Image of Better Response	
Description of Weaker Responses	*Weaker responses* showed misconceptions such as assuming calcium circulates freely without storage, not recognising bones as a reservoir and failed to explain deposition and release in response to changing blood calcium levels. Some students misunderstood that regulation involves maintaining balance, thinking it only adds calcium, while others confused it with unrelated processes. Overall, these responses lacked clarity, relevant scientific terms, and showed gaps in understanding calcium homeostasis.
Image of Weaker Responses	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts)	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

• Review past paper questions on the concept • Utilise the resource guide for additional materials	• Questioning Technique (Socratic approach) • Practical Demonstration	

Any Additional Suggestion: Use clear analogies such as comparing bones to a 'calcium bank' where minerals are deposited and withdrawn as needed to maintain blood calcium balance.

Question No. 3	
Question Text	Ambreen sees a snake crawling towards her. Write any TWO ways in which her sympathetic nervous system will respond to this stressful situation.
SLO No.	17.4.5
SLO Text	Differentiate between sympathetic and parasympathetic nervous system.
Max Marks	2
Cognitive Level	U
Checking Hints	1 mark for writing EACH way (any TWO required)
Overall Performance	Overall, candidates demonstrated a good understanding of the fight or flight responses mediated by the sympathetic nervous system. The use of appropriate terminology and clear, concise explanations reflected a strong grasp of the concept and its relevance to survival mechanisms.
Description of Better Responses	*Better responses* accurately described key effects of the sympathetic nervous system during the fight or flight response. Many candidates correctly mentioned that the heartbeat increases and the pupils dilate, showing understanding of how the body prepares for immediate action. These responses demonstrated clarity, appropriate use of scientific terms, and a good grasp of the physiological changes involved in stressful situations.
Image of Better Response	
Description of Weaker Responses	*Weaker responses* were vague and focused on behavioural actions, such as 'she uses a stick to beat the snake,' rather than physiological changes caused by the sympathetic nervous system. These answers lacked scientific explanation and did not address the body's internal responses like increased heart rate or pupil dilation. This suggests a misunderstanding of the concept and a need to distinguish between physiological responses and conscious actions in stressful situations.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion: Emphasise that the fight or flight response involves involuntary physiological changes, not deliberate actions or behaviours like hitting or running. Present situations (e.g., encountering a snake) and ask students to identify internal body changes, guiding them to focus on heart rate, pupil size, etc.

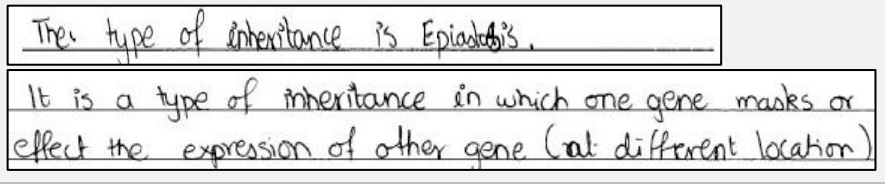
Question No. 4	
Question Text	Albinism in mice is caused by a gene that suppresses (masks) the expression of pigment in their eyes, skin, and hair. a. Identify the type of inheritance exemplified in the case of albinism in mice. b. Give a reason for identifying the type of inheritance in part 'a'.
SLO No.	22.4.4
SLO Text	Describe epistasis, dominant and recessive epistasis.
Max Marks	2
Cognitive Level	U
Checking Hints	1 mark for identification of type of inheritance 1 mark for giving the reason
Overall Performance	Overall performance on this question was poor. Candidates show limited grasp of genetic terminology and concepts. This reflects the need for better conceptual clarity and reinforcement of non-Mendelian inheritance patterns.
Description of Better Responses	*Better responses* correctly identified the type of inheritance as 'epistasis' and provided accurate reasoning that albinism occurs when one gene mask the expression of other genes located at different loci, preventing the production of pigment in eyes, skin, and hair. Their answers showed a clear understanding of gene interaction and the principle of epistasis, using appropriate terminology and logical descriptions.
Image of Better Response	
Description of Weaker Responses	*Weaker responses* revealed a lack of conceptual clarity regarding types of inheritance. Many candidates incorrectly identified the inheritance pattern as complete dominance, incomplete dominance, codominance, pleiotropy, or polygenic inheritance. These responses showed confusion between Mendelian and non-Mendelian inheritance patterns and a misunderstanding of how one gene can mask the expression of genes at other loci.

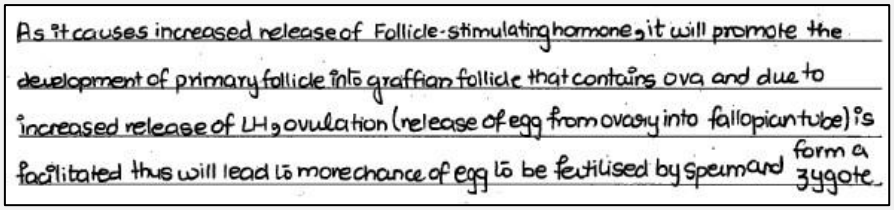
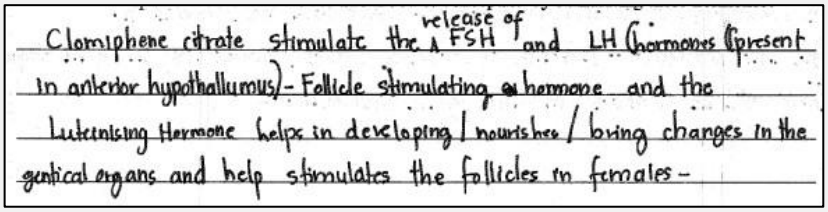
Image of Weaker Response	Complete dominance In complete dominance dominat gene express it self and mask the effect of the recessive gene.

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Pedagogy Used for that SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion: Use comparison charts to distinguish epistasis from complete dominance, incomplete dominance, codominance, pleiotropy, and polygenic inheritance. Additionally, implement hands-on activities like Punnett square exercises involving epistatic genes to reinforce understanding.

Question No. 5	
Question Text	Clomiphene citrate is a drug used to cure infertility in females. This drug causes increased release of FSH (Follicle Stimulating Hormone) and LH (Luteinising Hormone). How does clomiphene citrate enhance the likelihood of conception by stimulating these hormones?
SLO No.	18.6.4
SLO Text	Explain different stages of reproductive cycle and its hormonal control in human female.
Max Marks	2
Cognitive Level	U
Checking Hints	1 mark for the function of FSH 1 mark for the function of LH
Overall Performance	Overall, candidates performed well on this question. Many responses were well-structured and used appropriate biological terminology, reflecting a good grasp of hormonal regulation in the female reproductive system. This indicates effective teaching and sound understanding of the underlying physiological processes.
Description of Better Responses	*Better responses* demonstrated a clear understanding of how clomiphene citrate enhances the likelihood of conception by stimulating the release of FSH ((Follicle Stimulating Hormone) and LH. Candidates accurately explained that increased FSH promotes follicle development, while elevated LH (Luteinising Hormone) levels trigger ovulation, both of which are crucial for successful fertilisation.

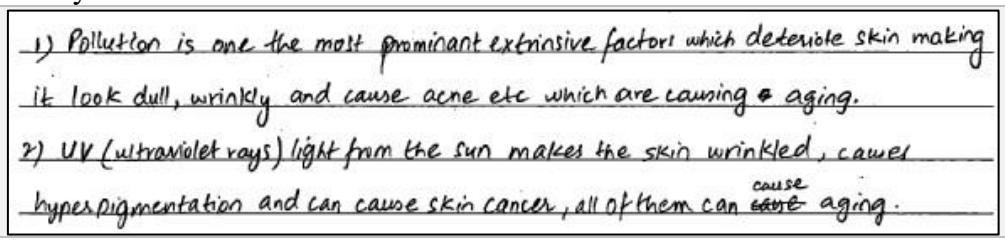
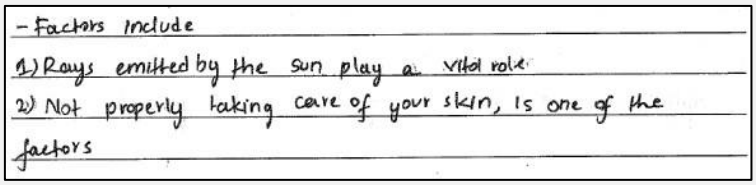
Image of Better Response	 As it causes increased release of Follicle-stimulating hormone, it will promote the development of primary follicle into graffian follicle that contains ova and due to increased release of LH, ovulation (release of egg from ovary into fallopian tube) is facilitated thus will lead to more chance of egg to be fertilised by sperm and form a zygote.
Description of Weaker Responses	*Weaker responses* lacked clarity and showed limited understanding of the roles of FSH and LH or provided vague explanations unrelated to hormonal stimulation. Some responses incorrectly focused on general fertility treatments without linking them to the hormonal mechanism involved. Others confused the functions of FSH and LH or omitted their impact on follicle development and ovulation.
Image of Weaker Response	 Clomiphene citrate stimulate the release of FSH and LH (hormones present in anterior hypothallumus) - Follicle stimulating hormone and the Luteinising Hormone helps in developing / nourishes / bring changes in the genital organs and help stimulates the follicles in females -

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Pedagogy Used for that SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion: Emphasise the specific functions of FSH and LH in the menstrual cycle using diagrams, animations, and flowcharts. Further, incorporate real-life or relatable scenarios involving women undergoing fertility treatment with certain drugs to help students connect theory with practical application.

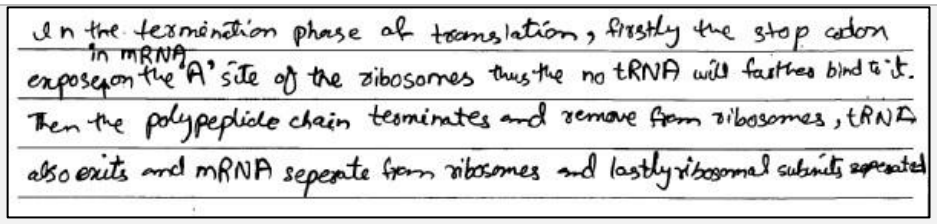
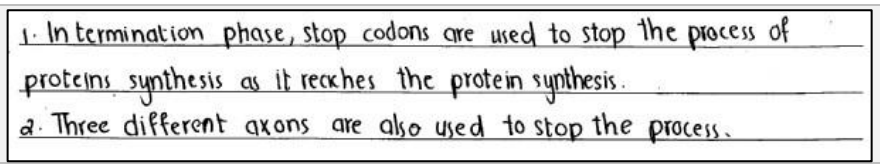
Question No. 6	
Question Text	State any TWO extrinsic factors that cause skin aging in humans.
SLO No.	19.6.5
SLO Text	list genetic and extrinsic factors responsible for aging.
Max Marks	2
Cognitive Level	K*
Checking Hints	1 mark for stating each factor (any TWO required) *(Marks will be awarded for stating any relevant factor.)*

Overall Performance	Overall, candidates performed well on this question. Most were able to correctly identify two extrinsic factors that contribute to skin ageing in humans. This suggests that the topic was well taught and that candidates were able to recall their knowledge effectively.
Description of Better Responses	*Better responses* accurately identified two valid extrinsic factors contributing to skin ageing. Many candidates correctly mentioned factors such as solar radiation, smoking, air pollution, and stress. Others also referred to climate conditions, poor nutrition, inadequate sleep, and lack of proper skincare. These answers demonstrated a well-rounded understanding of how lifestyle and environmental elements affect skin health.
Image of Better Response	
Description of Weaker Responses	*Weaker responses* were vague or imprecise, with some candidates giving unclear examples such as 'acid rain' or overly general statements like 'not taking proper skin care' without specifying the contributing factors. These answers lacked scientific accuracy and failed to clearly identify recognised extrinsic factors of skin ageing.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Pedagogy Used for that SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion: Present relatable scenarios (e.g., outdoor workers vs. office workers) to highlight lifestyle and environmental effects on skin. Clearly distinguish between intrinsic and extrinsic factors of ageing and provide students with an organised list of these factors to categorise and sort accordingly.

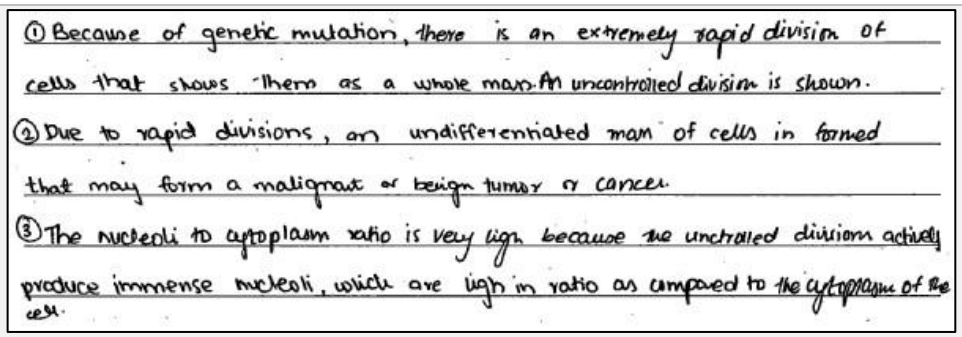
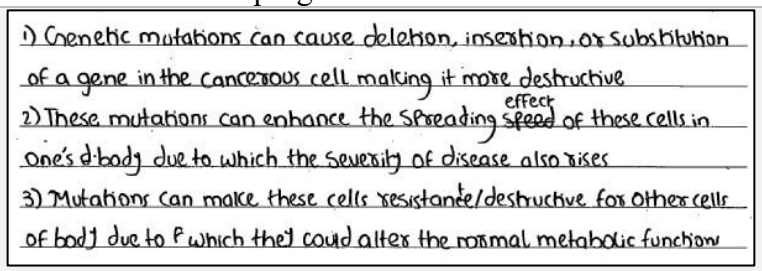
Question No. 7	
Question Text	Describe the termination phase of translation in the process of protein synthesis.
SLO No.	20.5.3
SLO Text	Explain mechanism of protein synthesis by means of DNA and RNA.
Max Marks	2
Cognitive Level	U
Checking Hints	1 mark for describing each step (any TWO required)
Overall Performance	The general performance of candidates on this question was average. While some responses showed a basic understanding of the termination phase of translation, many lacked details or omitted key steps. This indicates a partial understanding and suggests the need for reinforcing the specific events involved in the termination phase of translation in protein synthesis.
Description of Better Responses	*Better responses* accurately described the termination phase of translation in a clear and structured manner. Candidates correctly stated that termination begins when a stop codon (UAA, UAG, or UGA) enters the A site of ribosome and is recognised by release factors. They also explained that these factors trigger the addition of a water molecule, leading to the release of the polypeptide chain from the tRNA. Some responses further noted that the ribosome then dissociates into its subunits.
Image of Better Response	 In the termination phase of translation, firstly the stop codon in mRNA exposes on the 'A' site of the ribosomes thus the no tRNA will further bind to it. Then the polypeptide chain terminates and remove from ribosomes, tRNA also exits and mRNA seperate from ribosomes and lastly ribosomal subunits seperated
Description of Weaker Responses	*Weaker responses* revealed several common misconceptions about the termination phase of translation. Many candidates failed to mention the role of stop codons or incorrectly identified them as coding for amino acids. Some responses confused termination with transcription or other stages of translation, such as elongation. Several candidates did not mention release factors or misunderstood their function, often mentioning that tRNA brings the final amino acid. Others omitted the release of the polypeptide chain or the dissociation of ribosomal subunits.
Image of Weaker Response	 1. In termination phase, stop codons are used to stop the process of proteins synthesis as it reaches the protein synthesis. 2. Three different axons are also used to stop the process.

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Pedagogy Used for that SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion: Use step-by-step visuals to present the termination phase of translation using labelled diagrams or animations to illustrate the role of stop codons, release factors, and ribosome disassembly. Further, use guided practice by providing structured questions that guide students through each step of the termination process.

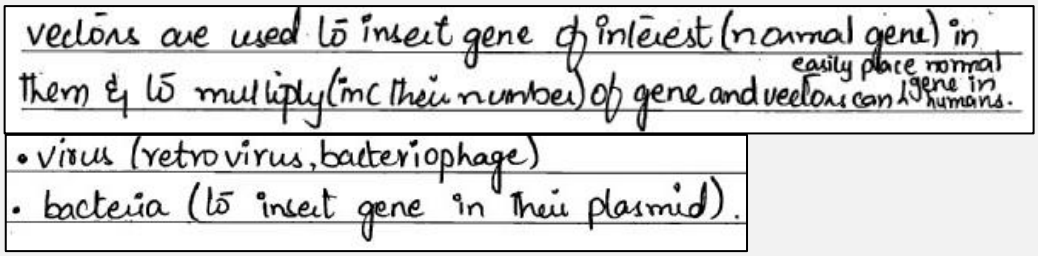
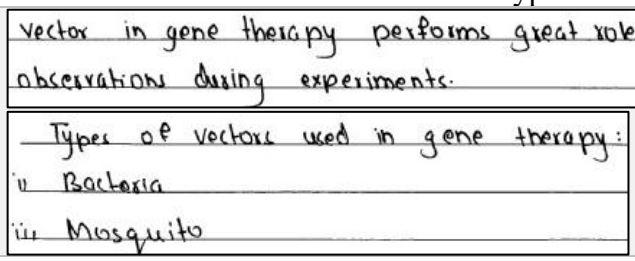
Question No. 8	
Question Text	Describe THREE basic changes in the metastatic cancerous cells that are caused by genetic mutations.
SLO No.	21.2.4
SLO Text	Describe cancer as uncontrolled cell division.
Max Marks	3
Cognitive Level	U
Checking Hints	1 mark for describing uncontrolled cell proliferation 1 mark for describing resistance to apoptosis 1 mark for describing metastasis
Overall Performance	The candidate performance in this question was good. Most candidates were able to describe three key changes in metastatic cancerous cells caused by genetic mutations. Responses reflected a solid understanding of the concept. Use of accurate scientific terminology and well-structured explanations indicated that the concept was well understood by most of the candidates.
Description of Better Responses	*Better responses* clearly described three key changes in metastatic cancerous cells caused by genetic mutations: uncontrolled proliferation, resistance to apoptosis, and the ability to metastasise. Some of them effectively distinguish malignant tumours from benign ones such as malignant tumours invade surrounding tissues and spread to distant sites, while benign tumours remain localised and do not spread. Many responses also highlighted typical features of cancerous cells, such as a high nucleus-to-cytoplasm ratio and loss of contact inhibition.

Image of Better Response	
Description of Weaker Responses	*Weaker responses* did not focus on the basic changes that occur in metastatic cancerous cells. Instead of describing key features, many candidates simply listed types of genetic mutations or provided irrelevant information. These responses lacked detail and did not address the cellular and behavioural changes that define malignant cancer. This suggests a misunderstanding of the question and indicates the need for clearer teaching on the specific characteristics and progression of metastatic cancerous cells.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Pedagogy Used for that SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login
Any Additional Suggestion: Use animations, models, or role-play activities to demonstrate how cancer cells differ from normal cells in terms of growth, apoptosis, and spread.		

Question No. 9	
Question Text	a. What is the function of vectors in gene therapy? b. Mention any TWO types of vectors commonly used in gene therapy.
SLO No.	23.4.3
SLO Text	Describe the process of gene therapy in the treatment of genetic diseases, i.e. a. Huntington's disease b. cystic fibrosis.
Max Marks	3

Cognitive Level	U
Checking Hints	1 mark for writing the function of vector in gene therapy 1 mark for mentioning the type of vector (any TWO required)
Overall Performance	Candidate performance on this question was generally moderate. These responses suggest a basic understanding of the topic but highlight the need for deeper clarification of vector function and types in gene therapy.
Description of Better Responses	*Better responses* accurately explained that vectors act as vehicles to deliver therapeutic genetic material, such as a functioning gene, directly into target cells. Candidates clearly articulated the role of vectors in gene therapy and provided correct examples, including modified viruses or bacteriophages, plasmids or bacteria, and liposomes.
Image of Better Response	
Description of Weaker Responses	*Weaker responses* revealed misconceptions about the function of vectors in gene therapy. Many candidates provided vague or incorrect explanations, failing to describe vectors as carriers of therapeutic genes into target cells. Some responses confused vectors with disease-carrying organisms, incorrectly mentioning examples like mosquitoes. Others listed unrelated tools or failed to name valid types of vectors altogether.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Pedagogy Used for that SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion: Incorporate real-world examples by discussing real gene therapy treatments (e.g., for SCID or cystic fibrosis) to contextualise how vectors function in medical applications.

Extended Response Questions (ERQs)

Extended response questions offered a choice between parts ‘a’ and ‘b’

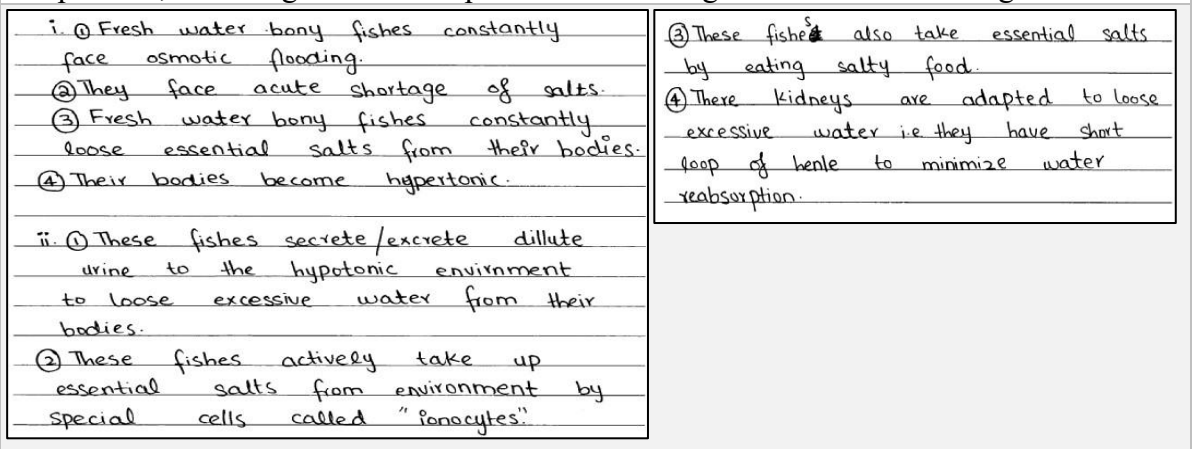
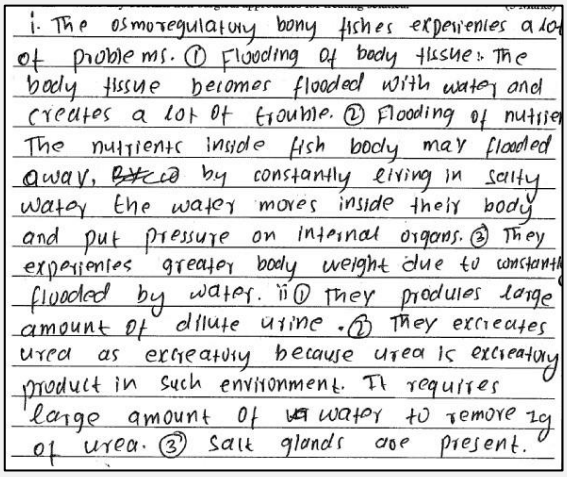
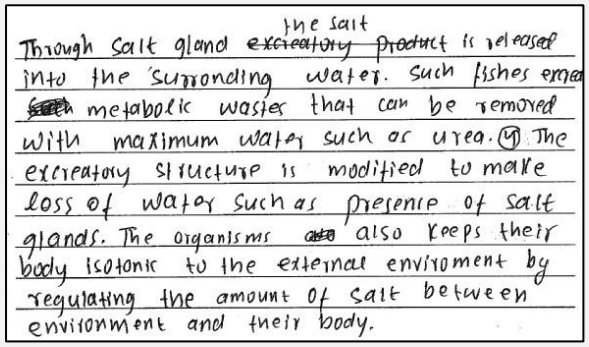
Question No. 10a	
Question Text	i. Identify THREE osmoregulatory challenges faced by freshwater bony fish. ii. Describe any FOUR osmoregulatory adaptations in freshwater bony fish to address these challenges.
SLO No.	15.3.4
SLO Text	Explain osmoregulation in aquatic (fresh water and marine) and terrestrial animals.
Max Marks	7
Cognitive Level	U
Checking Hints	i. 1 mark for identifying the hypotonic environment 1 mark for identifying the water influx 1 mark for identifying the ion loss ii. 1 mark for describing each adaptation (any FOUR required)
Overall Performance	Most of the candidates attempted this part of the question. Candidates showed average performance in this question. Some candidates were able to identify basic osmoregulatory challenges and name relevant adaptations. However, many responses lacked depth and clarity in linking specific adaptations to the challenges faced. Overall, the question was attempted with average performance, but students need to strengthen their understanding of the physiological mechanisms involved in freshwater osmoregulation.
Description of Better Responses	*Better responses* demonstrated a clear understanding of the osmotic challenges in a hypotonic freshwater environment, correctly identifying water influx and ion loss as key issues. These candidates accurately described physiological adaptations such as dilute urine production, active ion uptake via gill ionocytes, and absorption of salts from food. They also recognised that freshwater fish do not drink water and explained why. Responses showed good use of scientific terminology and logical links between challenges and adaptations, reflecting sound conceptual understanding of freshwater osmoregulation.
Image of Better Response	
Description of Weaker Responses	*Weaker responses* showed confusion between freshwater and marine adaptations, with some incorrectly stating that freshwater fish drink water or face dehydration. Misuse of terminology was common, including references to ‘lungs’ instead of gills and incorrect statements like salts being absorbed through kidneys or excreted via skin. Additionally, several candidates gave incomplete descriptions of adaptations, such as mentioning ‘active transport’ without identifying ionocytes or the gills as the site of action. These inaccuracies indicate a need for clearer understanding of fish physiology and osmoregulatory mechanisms specific to freshwater environments.

Image of Weaker Response



i. The osmoregulatory bony fishes experiences a lot of problems. ① Flooding of body tissue: The body tissue becomes flooded with water and creates a lot of trouble. ② Flooding of nutrie The nutrients inside fish body may flooded away, ~~B~~ by constantly living in salty water the water moves inside their body and put pressure on internal organs. ③ They experiences greater body weight due to constant flooded by water. ii ① They produces large amount of dilute urine. ② They excreates urea as excreatory because urea is excreatory product in such environment. It requires large amount of ~~wa~~ water to remove 1g of urea. ③ Salt glands are present.

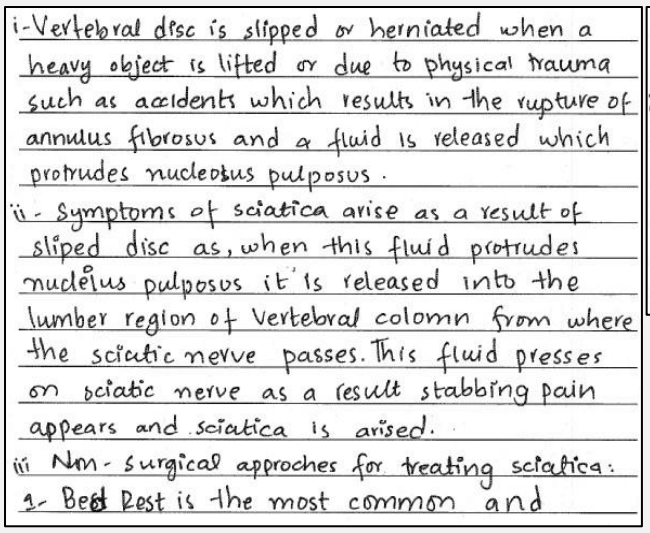
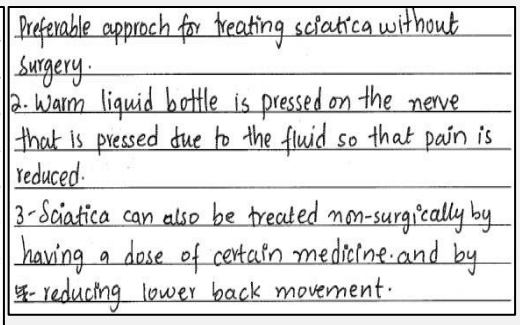


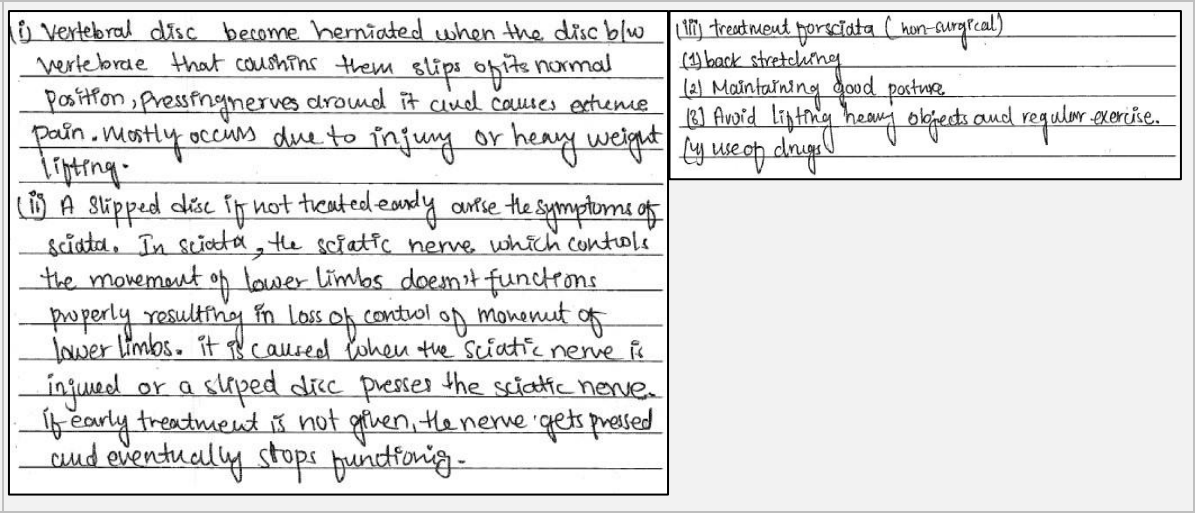
Through salt gland the salt ~~excreatory product~~ is released into the surrounding water. Such fishes ~~exc~~ metabolic wastes that can be removed with maximum water such as urea. ④ The excreatory structure is modified to make loss of water such as presence of salt glands. The organisms ~~are~~ also keeps their body isotonic to the external environment by regulating the amount of salt between environment and their body.

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

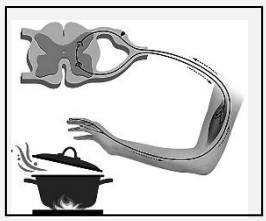
Any Additional Suggestion: Emphasise the contrast between freshwater and marine osmoregulation, as well as the functional link between the identified challenges and physiological adaptations. Visual aids like diagrams of freshwater fish showing gill function, kidney structure, and urine output could also help students develop a more integrated understanding.

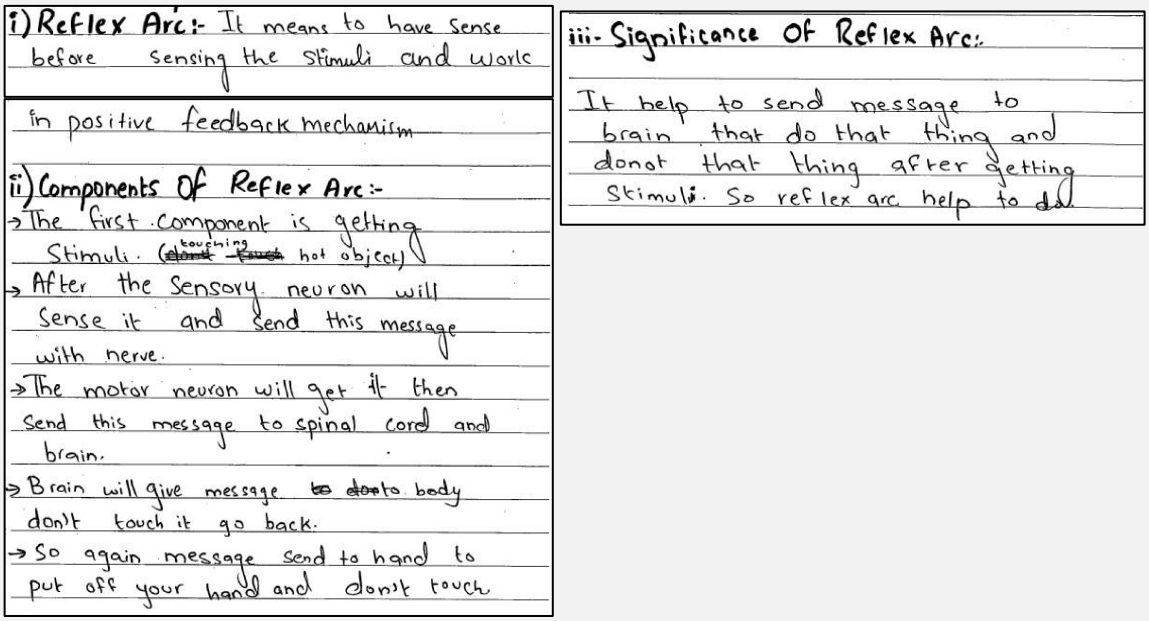
Question No. 10b	
Question Text	i. How does a vertebral disc become slipped or herniated? ii. Describe how the symptoms of sciatica arise as a result of slipped discs in the lumbar (lower back) region. iii. Write any THREE non-surgical approaches for treating sciatica.
SLO No.	16.4.8
SLO Text	Discuss skeleton related diseases and their preventive measures, i.e. a. disc slip b. spondylitis c. sciatica d. osteoarthritis.
Max Marks	7
Cognitive Level	U
Checking Hints	i. 1 mark for explaining degeneration and wear as a cause for disc weakening 1 mark for describing how excessive pressure or strain can lead to the bulging or herniation of the nucleus pulposus ii. 1 mark for describing disc compression on spinal nerves 1 mark for describing the symptoms (any 1) iii. 1 mark for writing each non-surgical treatment (any THREE required)
Overall Performance	This part of the question was attempted less frequently compared to part (a). However, those who attempted it showed average performance. Many candidates demonstrated a basic understanding of slipped discs and sciatica but lacked depth in their explanations. Greater focus on anatomical understanding and symptom-mechanism relationships is recommended.
Description of Better Responses	*Better responses* clearly explained that slipped discs result from degeneration of the annulus fibrosus and excessive pressure causing the nucleus pulposus to protrude. Candidates accurately linked sciatic symptoms to compression of the sciatic nerve in the lumbar region, describing radiating pain, tingling, and numbness in the leg. Responses also demonstrated good understanding of non-surgical treatments such as physiotherapy, hot/cold packs, medication, and steroid injections. Use of appropriate anatomical terms and logical explanation of symptom development and treatment approaches reflected a sound grasp of the topic.
Image of Better Response	 i-Vertebral disc is slipped or herniated when a heavy object is lifted or due to physical trauma such as accidents which results in the rupture of annulus fibrosus and a fluid is released which protrudes nucleosus pulposus. ii- Symptoms of sciatica arise as a result of sliped disc as, when this fluid protrudes nucleus pulposus it is released into the lumber region of vertebral colomn from where the sciatic nerve passes. This fluid presses on sciatic nerve as a result stabbing pain appears and sciatica is arised. iii Non-surgical approches for treating sciatica: 1- Best Rest is the most common and  Preferable approch for treating sciatica without surgery. 2. Warm liquid bottle is pressed on the nerve that is pressed due to the fluid so that pain is reduced. 3-Sciatica can also be treated non-surgically by having a dose of certain medicine and by reducing lower back movement.
Description of Weaker Responses	*Weaker responses* were often overly generalised, attributing slipped discs to vague reasons like 'external injuries' without referencing specific anatomical changes such as degeneration of the annulus fibrosus. Some incorrectly described stiffness in bones rather than joints, indicating confusion in basic musculoskeletal concepts. In part (iii), many candidates hardly listed 'exercise' as a treatment without elaborating on types or other non-surgical options like medication or physiotherapy. These responses lacked clarity,

	specificity, and depth, showing a limited understanding of the causes, symptoms, and management of sciatica related to lumbar disc herniation.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login
Any Additional Suggestion: Create side-by-side comparisons of healthy vs. herniated discs and sciatica symptoms vs. general lower back pain to deepen conceptual clarity.		

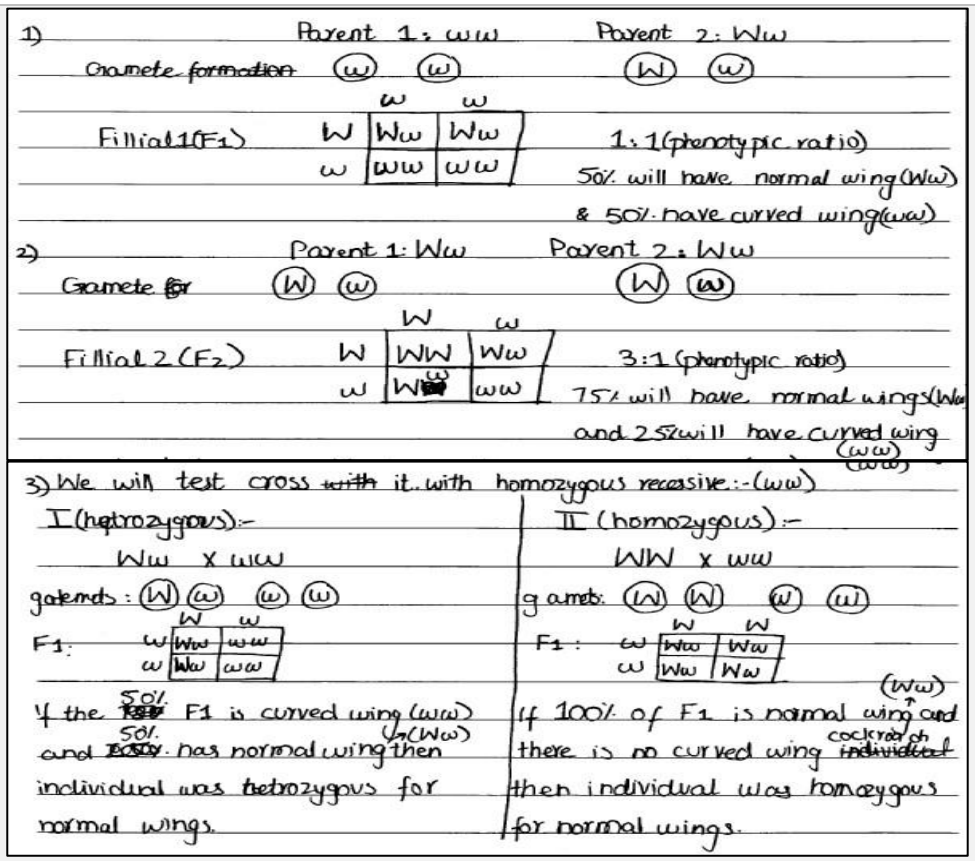
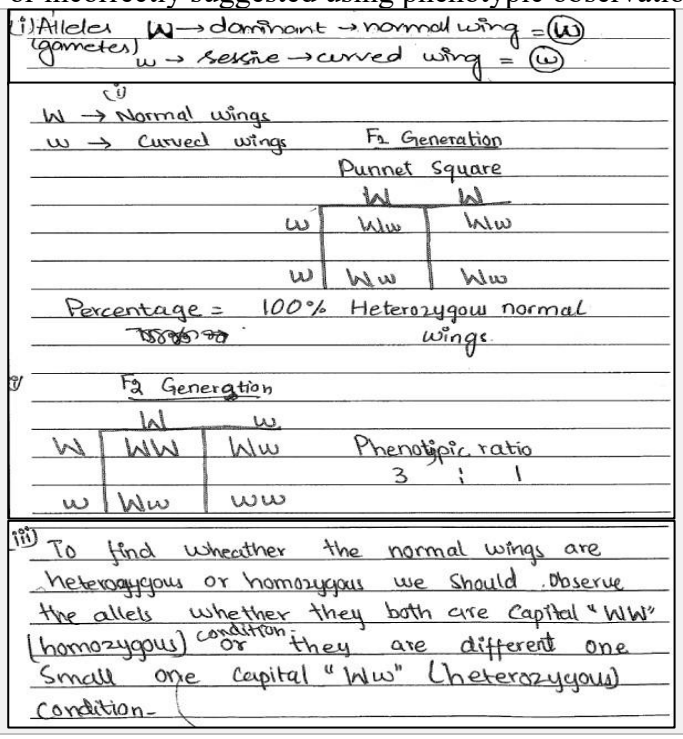
Question No. 11a	
Question Text	The given diagram shows a simple reflex arc.  i. What is a reflex arc? ii. With reference to the components of reflex arc, describe the sequential steps of the shown reflex arc. iii. Write any ONE point of significance of this reflex arc.
SLO No.	17.3.5
SLO Text	Explain reflex arc.
Max Marks	7
Cognitive Level	U
Checking Hints	1 mark for defining reflex arc 1 mark for describing each component with its action (FIVE required) 1 mark for writing the significance (any ONE required)
Overall Performance	Most of the candidates were inclined towards this part of the question, likely because the concept is also covered in Grade 10, making it familiar and accessible. Overall, responses were satisfactory with room for improvement in detail and terminology.
Description of Better Responses	*Better responses* clearly defined a reflex arc as an automatic, rapid, and involuntary response to a stimulus. These candidates accurately identified and described the components such as receptor, sensory neuron, relay neuron, motor neuron, and effector in the correct sequence. They demonstrated a strong understanding of the pathway by logically linking each component to the next. For part (iii), better answers effectively explained the significance of reflex arcs in protecting the body from harm by enabling quick responses without involving the brain.
Image of Better Response	a) i) Reflex arc is the pathway followed by reflex action (it is the immediate and involuntary response). ii) a person touched the hot lid and immediately withdrew his hands from it, his skin receptors that are thermoreceptors detected this stimuli and ~~send~~ sent the signal to sensory neuron through sensory neuron it travelled to the central center that is spinal cord (as it is a shock absorber) there lies an inter neuron, through the synapse the message travelled towards motor neuron and in spinal cord the information was processed and now the motor neuron is carrying the signal to the effector that will help in the response that what the person will do, an effector in this case is muscle as the motor neuron reached the muscles it contracted withdrawing the hand away from the lid, this happens within ~~xxxx~~ seconds. iii) Significance of reflex arc is that as it is immediate response it helps the person from damage and injury.
Description of Weaker Responses	*Weaker responses* often showed confusion about the definition and function of a reflex arc, with some describing it simply as 'a body movement' or 'an action done by the brain,' missing the key feature of it being an involuntary response. Common misconceptions

	included mixing up the sequence of components (e.g., placing the motor neuron before the sensory neuron) or omitting key parts such as the relay neuron. Some candidates incorrectly stated that the brain controls the reflex action, rather than recognising it as a spinal cord response. A few also gave vague or unrelated points for the significance of the reflex arc.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Pedagogy Used for that SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login
Any Additional Suggestion: Emphasise that reflex arcs are automatic and spinal cord-based, not controlled by the brain, to correct common misconceptions. Additionally, incorporate step-by-step diagrams and flowcharts showing the correct sequence of reflex arc components to help students visualise the process.		

Question No. 11b	
Question Text	In cockroach (*Periplaneta americana*), the trait of normal-shaped wings is dominant over the curved-shape wings, which is controlled by a single gene with a pair of alleles. i. Construct a Punnet square to determine the phenotype percentage of offspring when a homozygous individual with curved wings was allowed to breed with a heterozygous individual. ii. Deduce the phenotype ratio of the F2 generation if the offspring with normal shape wings (of cross done in part 'i') were allowed to interbreed together. iii. Describe how one could find out whether individuals were homozygous or heterozygous for the normal wings. (**Note**: Use the symbol 'W' to represent the dominant allele for normal wings and 'w' to represent the recessive allele for curved wings.)
SLO No.	22.2.4
SLO Text	Illustrate Mendel's laws through genetic crosses.
Max Marks	7
Cognitive Level	A*
Checking Hints	i. 1 mark for placing the correct gametes in Punnet square to cross 1 mark for mentioning the genotypes of offspring in the Punnet square 1 mark for mentioning the phenotype percentage of offspring in Punnet square ii. 1 mark for deducing the phenotype ratio of F2 generation iii. 1 mark for writing test cross of normal wings with the homozygous recessive curved wings 1 mark for writing if all offspring are with normal wings, then test parent is probably WW/homozygous 1 mark for writing if some offspring are with curved wings, then test parent is Ww/heterozygous
Overall Performance	This part of the question was attempted by a smaller number of candidates; however, those who did attempt it generally performed well. Their responses showed a solid understanding of Mendelian inheritance. The correct use of genetic symbols and clear, logical explanations indicated a strong command of fundamental genetic concepts and their application to problem-solving.
Description of Better Responses	*Better responses* demonstrated accurate application of Mendelian genetics. Candidates correctly constructed the Punnett square using appropriate genetic symbols (W and w), and accurately calculated phenotype percentages for the first cross. In part (ii), they logically deduced the F2 generation's phenotype ratio, showing clear understanding of monohybrid inheritance. For part (iii), better answers clearly explained the concept of a test cross using a homozygous recessive individual to identify unknown genotypes. Overall, responses reflected strong analytical skills, correct terminology, and clear reasoning.

Image of Better Response	
Description of Weaker Responses	*Weaker responses* revealed several common misconceptions in applying Mendelian genetics. Some candidates incorrectly used the same letter for both dominant and recessive alleles (e.g., W and w), while others reversed the meaning of capital and lowercase letters. In Punnett squares, a few showed all offspring as having curved wings despite one parent being heterozygous. In part (ii), some believed that only normal-winged offspring could be produced from two normal-winged parents, overlooking the possibility of heterozygous crosses producing recessive traits. In part (iii), many confused a test cross with self-crossing or incorrectly suggested using phenotypic observation alone to determine genotype.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login
Any Additional Suggestion: Use real-life contexts by applying genetic problems to familiar organisms or traits to make abstract concepts more relatable and engaging for students.		

Annexure A: Pedagogies Used for Teaching the SLOs

Pedagogy: Storyboard
Description: A visual pedagogy that uses a series of illustrated panels to present a narrative, encouraging creativity and critical thinking. It helps learners organise ideas, sequence events, and comprehend complex concepts through storytelling.
Example: In a Literature class, students are tasked with creating storyboards to visually retell a novel. They draw key scenes, write captions, and present their stories to the class, enhancing their reading comprehension and fostering their imagination.

Pedagogy: Cause and Effect
Description: This pedagogy explores the relationships between actions and consequences. By analysing cause-and-effect relationships, learners develop a deeper understanding of how events are interconnected and how one action can lead to various outcomes.
Example: In a History class, students study the causes and effects of the Industrial Revolution. They research and discuss how technological advancements in manufacturing led to significant societal changes, such as urbanisation and labour reform movements.

Pedagogy: Fish and Bone
Description: A method that breaks down complex topics into main ideas (the fish) and supporting details (the bones). This visual approach enhances comprehension by highlighting essential concepts and their relevant explanations.
Example: During a Biology class on human anatomy, the teacher uses the fish and bone technique to teach about the human skeletal system. Teacher presents the main components of the human skeleton (fish) and elaborates on each bone's structure and function (bones).

Pedagogy: Concept Mapping
Description: An effective way to visually represent relationships between ideas. Learners create diagrams connecting key concepts, aiding in understanding the overall structure of a subject and fostering retention.
Example: In a Psychology assignment, students use concept mapping to explore the various theories of personality. They interlink different theories, such as Freud's psychoanalysis, Jung's analytical psychology, and Bandura's social-cognitive theory, to see how they relate to each other.

Pedagogy: Audio Visual Resources
Description: Incorporating multimedia elements like videos, images, and audio into lessons. This approach caters to different learning styles, making educational content more engaging and memorable.
Example: In a General Science class, the teacher uses a documentary-style video to teach about the solar system. The video includes stunning visual animations of the planets, interviews with astronomers, and background music, enhancing students' interest and understanding of space.

Pedagogy: Think, Pair, and Share
Description: A collaborative learning technique where students ponder a question or problem individually, then discuss their thoughts in pairs or small groups before sharing with the entire class. It fosters active participation, communication skills, and diverse perspectives.
Example: In a Literature in English class, the teacher poses a thought-provoking question about a novel's moral dilemma. Students first reflect individually, then pair up to exchange their opinions, and finally participate in a lively class discussion to explore different viewpoints.

Pedagogy: Questioning Technique (Socratic Approach)
Description: Based on Socratic dialogue, this method stimulates critical thinking by posing thought-provoking questions. It encourages learners to explore ideas, justify their reasoning, and discover knowledge through a process of inquiry.
Example: In an Ethics class, the instructor uses the Socratic approach to lead a discussion on the meaning of justice. By asking a series of probing questions, the students engage in a deeper exploration of ethical principles and societal values.

Pedagogy: Practical Demonstration
Description: A hands-on approach where learners observe real-life applications of theories or skills. Practical demonstrations enhance comprehension, skill acquisition, and problem-solving abilities by bridging theoretical concepts with real-world scenarios.
Example: In a Food and Nutrition class, the instructor demonstrates the proper technique for filleting a fish. Students observe and then practice the skill themselves, learning the practical application of knife skills and culinary precision.

(**Note:** The examples provided in this annexure serve as illustrations of various pedagogies. It is important to understand that these pedagogies are versatile and can be applied across subjects in numerous ways. Feel free to adapt and explore these techniques creatively to enhance learning outcomes in your specific context.)

Acknowledgements

The Aga Khan University Examination Board (AKU-EB) acknowledges with gratitude the invaluable contributions of all the dedicated individuals who have played a pivotal role in the development of the Biology HSSC-II E-Marking Notes.

We extend our sincere appreciation to Sajida Mohammed Afzal, Specialist in Biology at AKU-EB, and Dr Nazish George, Specialist in Biology at AKU-EB, for taking subject lead during the entire process of e-marking.

Additionally, we express our gratitude to the esteemed team of reviewers for their constructive feedback on overall performance, better and weaker responses, and validating teaching pedagogies along with suggestions for improvement.

These contributors include:

• Zain Muluk, Manager, Examination Development, AKU-EB

• Munira Mohammad, Manager, Assessment, AKU-EB

• Dr Naveed Yousuf, CEO, AKU-EB